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Monthly Progress Report – Month 1 (2026)

Project Overview

Project Title: AI-Based Educational Content Viewer with Topic Summarization

This project is being carried out at Krishna Kanta Handiqui State Open University (KKHSOU) as part of the MCA 4th Semester Industrial Project. The goal of the project is to develop an intelligent system that allows users to read academic PDF materials through an interactive interface and automatically generate summaries of the content using Artificial Intelligence techniques.

Objectives of the Project: • Develop an interactive PDF viewer for educational documents. • Implement an AI-based system to generate summaries of document content. • Improve accessibility and readability of digital learning materials. • Understand the application of AI technologies in educational platforms.

Technologies Used: • Python • Flask (Web framework) • PyMuPDF for PDF processing • NLP / Transformer-based models for summarization • HTML, CSS, JavaScript for the user interface

Organization Profile: Krishna Kanta Handiqui State Open University (KKHSOU) is a state university in Assam that provides higher education through open and distance learning. The university uses digital technologies to deliver educational content and support learning for students.

Work Done in Month 1

• Studied the project objectives and scope related to digital learning tools. • Conducted a literature review on AI-based text summarization techniques. • Explored methods for extracting text from PDF documents using Python libraries. • Installed and configured the required development environment and libraries. • Learned the basics of Natural Language Processing (NLP) used for summarizing textual data. • Designed an initial conceptual architecture for the system. • Began developing the basic web application structure using Flask. • Explored interactive ways to display PDF content using a flipbook-style interface.

Challenges Faced:

1. Processing Text from PDF Documents Extracting structured text from PDF files required experimenting with different libraries. Resolution: PyMuPDF was selected for efficient text extraction.

2. Understanding NLP Summarization Models Understanding how AI models summarize long text content required additional study. Resolution: Reviewed tutorials and documentation on transformer-based models.

Month 1 Milestones

Requirement Analysis – Completed (100%) Literature Review – Completed (100%) Development Environment Setup – Completed (100%) Initial System Architecture – 80% Completed

Next Month Plan

• Finalize the system architecture. • Implement the PDF viewing interface. • Design page navigation and content extraction features. • Develop backend logic for document processing. • Prepare the system for integration with the AI summarization module.

Conclusion

The first month of the project focused on understanding requirements, studying relevant technologies, and preparing the development environment. Literature review, environment setup, and initial system design were completed successfully. These steps created a strong foundation for developing the AI-based educational content viewer and summarization system in the upcoming months.

Bibliography

• Documentation of Python libraries used for PDF processing • Research papers on Natural Language Processing and text summarization • Official Flask documentation • Online tutorials and technical resources related to AI and NLP